1 Mocks

2 Pipeline choices

This section lists every choice made in the lens-modelling pipeline, stage by stage. It describes the real-data pipeline; Section 2.8 lists what differs for the mock data. The pipeline is a Source, Light and Mass (SLaM) chain fitted with `PyAutoLens` [Nightingale et al., 2021], followed by the subhalo search of Nightingale et al. [2024]. Every stage is sampled with the nested sampler `Nautilus` [Lange, 2023].

2.1 General set-up

Joint multi-band fitting. Each waveband is one factor of a factor graph and all bands are fitted simultaneously by one non-linear search per stage. A prior object placed in several bands' models is a single parameter of the joint fit. The same code fits one band or several, so the single-band and joint-band arms of the experiment differ only in their data.

- **Shared by all bands:** the lens mass (`Isothermal`, later a `PowerLaw`), the external shear and the dark matter subhalo, because lensing is achromatic.
- **Per band:** the lens light (its own multi-Gaussian expansion, MGE, with its own centre and ellipticity, because the stellar light need not have the same centroid or shape at every wavelength), the source (an MGE, later a pixelized source with its own mesh and regularization), the adapt images, the over-sampling, and the band's alignment to the reference band.

Reference band and alignment. The first band given is the astrometric reference. Its alignment is fixed (zero offset, zero rotation). Every other band has three free alignment parameters in every stage in which the alignment is free: a (y, x) grid offset, each with a uniform prior on $\pm 0.5''$, and a grid rotation angle with a uniform prior on $\pm 10°$. The rotation is about the offset point. The background sky level is fixed at zero in every band. Alignment can be fixed for all bands with `--fixed_alignment`.

Prior chaining. When a later stage takes a component from an earlier one as a *model* (free), the installed `PyAutoFit` reuses the earlier stage's *original* prior unchanged; the prior is not narrowed by the earlier posterior.

A component taken as an *instance* is fixed at the earlier stage's maximum-likelihood value. The linear light profiles of a fixed lens-light MGE keep their geometry (centre, ellipticity), but their amplitudes are still solved linearly in every likelihood evaluation.

Data preparation. Each band is loaded with its own pixel scale and given a circular mask of radius $3.5''$ (the `mask_radius` of the band's `info.json`), centred on the band's `mask_centre` (default $(0'', 0'')$). Light profiles are oversampled in radial bins about the mask centre: sub-grid size 4 within $0.3''$, 2 between $0.3''$ and $0.6''$, and 2 beyond. Pixelized sources use the sparse (w-tilde) inversion operator to fit in GPU memory; it gives the same likelihood as the dense formalism. For real data, every band is plotted and inspected for contaminating galaxies, stars and artefacts before fitting, and its mask is set from that inspection.

Position penalty. The multiple-image positions of the lensed source are supplied in the reference band's frame. The penalty is attached to the reference band's factor only. It constrains the shared mass model, so one copy suffices, and a copy per band would penalize single- and multi-band fits differently. Where the threshold is computed from an earlier result, it is twice the largest source-plane separation of the positions traced through that result's maximum-likelihood model, clipped to $[0.2'', 0.8'']$.

Sampler settings. Every search uses `Nautilus` on one GPU with `n_batch` = 10 (the command-line default, applied to every stage; `--n_batch` overrides it) and one core. All other `Nautilus` settings are the library defaults (`enlarge_per_dim` = 1.1, `split_threshold` = 100, `n_networks` = 4, `f_live` = 0.01, `n_eff` = 500, `n_shell` = 1, no fixed seed). JAX runs in double precision. The number of live points grows by a fixed increment for each band beyond the first, scaled to the parameters each band adds to that stage. Early stages are capped with `n_like_max`; the no-subhalo baseline and the subhalo searches are not, because a search stopped before convergence returns a biased evidence.

Table 1 summarizes the stages; N is the number of bands.

2.2 `source_lp[1]`: parametric source

The lens light, lens mass and a parametric source are fitted together.

Table 1: Stages of the real-data pipeline, for N bands. Free parameters are given for one and for two bands. Every stage uses `n_batch` $= 10$. The position-penalty threshold is either fixed or computed from `source_lp[1]`.

Stage	Free (1 / 2)	n_{live}	$n_{\text{like,max}}$	Pos. threshold
`source_lp[1]`	17 / 30	$200 + 80(N-1)$	2×10^5	fixed, $0.3''$
`source_pix[1]`	10 / 16	$150 + 50(N-1)$	4×10^5	computed
`source_pix[2]`	3 / 6	$75 + 25(N-1)$	5×10^4	none
`light[1]`	6 / 12	$300 + 100(N-1)$	2×10^5	none
`mass_total[1]`	8 / 11	$200 + 50(N-1)$	none	computed
`subhalo[1]`, per cell	11 / 14	600	none	computed
`subhalo[2]`	11 / 14	600	none	computed

- **Lens mass (shared):** an elliptical `Isothermal`. Centre: Gaussian priors, mean $0''$, $\sigma = 0.1''$. Elliptical components: truncated Gaussian priors, mean 0, $\sigma = 0.3$, truncated to $[-1, 1]$. Einstein radius: uniform on $[0'', 3.5'']$ (zero to the mask radius). The mass centre is free and is not tied to any band's lens-light centre.

- **External shear (shared):** γ_1 and γ_2 uniform on $[-0.3, 0.3]$.

- **Lens light (per band):** an MGE of two bases of 30 linear Gaussians each. The Gaussian widths are fixed and log-spaced from one tenth of the band's pixel scale to the mask radius. Both bases share one centre, with uniform priors of $\pm 0.1''$ about the band's mask centre. Each basis has its own elliptical components, with uniform priors on $[-0.5, 0.5]$. This gives six non-linear parameters per band.

- **Source (per band):** an MGE of one basis of 30 linear Gaussians, with widths log-spaced from one tenth of the pixel scale to $1.0''$. Centre: Gaussian priors on the mask centre with $\sigma = 0.6''$. Elliptical components: truncated Gaussian priors as for the mass. This gives four non-linear parameters per band.

- **Alignment:** free for every non-reference band, with the priors of Section 2.1.

- **Position penalty:** a fixed source-plane threshold of $0.3''$.

A separate lens-light stage is not run first, because the lens is a single isolated galaxy.

2.3 `source_pix[1]`: first pixelized source

Each band's parametric source is replaced by a pixelized source.

- **Source (per band):** a Delaunay mesh whose image-plane vertices come from a Hilbert image mesh. The Hilbert mesh has 1500 points at a $0.03''$ pixel scale and 1250 at $0.06''$, with a weight power of 3.5 and a weight floor of 0.01. Its adapt data is the band's lens-subtracted source signal-to-noise map from `source_lp[1]`, uncapped. Thirty further points are placed on a circle just outside the mask (radius: mask radius plus half a pixel), and the 30 Delaunay pixels they form are zeroed so that they carry no source flux.
- **Regularization (per band):** `AdaptSplit`, driven by the same signal-to-noise map. Inner and outer coefficients: log-uniform priors on $[10^{-6}, 10^{6}]$. Signal scale: uniform on $[0, 1]$. This gives three parameters per band.
- **Over-sampling of the pixelized source (per band):** sub-grid size 4 in pixels whose source signal-to-noise exceeds 3, and 2 elsewhere.
- **Lens mass (shared):** `Isothermal`, free with the `source_lp[1]` priors.
- **External shear (shared):** free with the `source_lp[1]` priors.
- **Lens light (per band):** fixed at its `source_lp[1]` maximum-likelihood geometry.
- **Alignment:** free with the `source_lp[1]` priors.
- **Position penalty:** threshold computed from `source_lp[1]`.

2.4 `source_pix[2]`: refined pixelized source

Each band's image mesh, adapt images and pixelization over-sampling are rebuilt from its `source_pix[1]` reconstruction. The adapt data fed to the image mesh is capped at a signal-to-noise of 3, so that the brightest source pixels do not take most of the mesh. The adapt images used by the regularization and the over-sampling remain uncapped. The datasets and adapt images built here are reused by every later stage.

- **Free:** only each band's `AdaptSplit` parameters, with the priors above.

- **Fixed:** the lens mass, shear and alignment at their `source_pix[1]` maximum-likelihood values, and the lens light at its `source_lp[1]` values.
- **Position penalty:** none.

The bands share no free parameter, so this joint fit is equivalent to independent per-band fits.

2.5 `light[1]`: lens light

The lens light is refitted against the pixelized sources.

- **Lens light (per band):** a fresh MGE of two bases of 30 linear Gaussians, with the same widths as in `source_lp[1]`. Centre: uniform priors of $\pm 0.1''$ about that band's `source_lp[1]` lens-light centre. Elliptical components: truncated Gaussian priors, mean 0, $\sigma = 0.3$, truncated to $[-1, 1]$; these differ from the uniform ± 0.5 priors of `source_lp[1]`.
- **Fixed:** each band's pixelized source (mesh and regularization) from `source_pix[2]`, and the lens mass, shear and alignment from `source_pix[1]`.
- **Position penalty:** none.

The bands again share no free parameter.

2.6 `mass_total[1]`: total mass and the no-subhalo baseline

The shared lens mass becomes an elliptical `PowerLaw`.

- **Lens mass (shared):** `PowerLaw`. Its centre, elliptical components and Einstein radius take the `source_pix[1]` `Isothermal` priors. Density slope: uniform on $[1.5, 3.0]$.
- **External shear (shared):** free with its original priors.
- **Alignment:** free with its original priors.
- **Lens light (per band):** fixed from `light[1]`.
- **Source (per band):** fixed from `source_pix[2]` (mesh, image-plane mesh and regularization).
- **Position penalty:** threshold computed from `source_lp[1]`.

- **Sampling:** uncapped, because the evidence of this fit is the no-subhalo baseline of the detection.

This fit is not repeated before the subhalo search. A refit without a subhalo, on the same analyses and with the same priors, would reproduce it, because chaining reuses the original priors.

2.7 Subhalo search

The two subhalo stages compose their models from `mass_total[1]`. The lens mass, shear and alignment are free with the same priors as in `mass_total[1]`. The lens light and pixelized source of each band are fixed at their `mass_total[1]` values. The fits run on the very analysis objects used by `mass_total[1]`: the same datasets, adapt images and position penalty. They therefore differ from the baseline only by one subhalo, shared by all bands.

Subhalo model. A spherical NFW halo at the lens redshift, whose concentration follows the mass–concentration relation of Ludlow et al. [2016] (`NFWMCRLudlowSph`). Its only mass parameter is M_{200}, with a log-uniform prior on $[10^6, 10^{11}]\,\mathrm{M}_\odot$. The lens and source redshifts in the profile are fixed to those of the system.

`subhalo[1]`: grid search. The subhalo centre is searched on a 2×2 grid over $(-3'', 3'')$ in y and x. Each cell is a separate joint fit of all bands, with uniform priors on its $3'' \times 3''$ cell for the centre. Nightingale et al. [2024] used 5×5 over the same region; 2×2 is chosen to cut the cost of the grid. The cells run one after another on one GPU. The grid can be changed with `--grid_steps` and `--grid_dimension`.

`subhalo[2]`: refinement. The subhalo is refitted with its centre free across the image plane. The centre's priors are Gaussian, centred on the maximum-likelihood subhalo centre of the highest-*evidence* grid cell, with σ equal to half a cell width ($1.5''$ for the 2×2 grid). M_{200} keeps its full log-uniform prior. The best cell is chosen by evidence rather than likelihood, because the evidence is the detection statistic.

Detection statistic. Two increases relative to `mass_total[1]` are reported:

$$\Delta \ln Z_{\mathrm{grid}} = \ln Z_{\text{best cell}} - \ln Z_{\texttt{mass_total[1]}}, \tag{1}$$

$$\Delta \ln Z_{\mathrm{refine}} = \ln Z_{\texttt{subhalo[2]}} - \ln Z_{\texttt{mass_total[1]}}. \tag{2}$$

Both depend on the centre prior through its Occam factor. A $3'' \times 3''$ cell is $\ln 4 \approx 1.4$ denser at the subhalo than a uniform prior over the whole $6'' \times 6''$ region. Taking the best of N_{cells} cells is a look-elsewhere selection, so $\Delta \ln Z_{\mathrm{grid}} - \ln N_{\mathrm{cells}}$ approximates the evidence for a subhalo whose position is unknown. The refined value is typically about 0.5 below the grid value, but its prior is centred using the same data, which favours the subhalo model. The two are read together. Absolute evidences of different band sets are never compared, because they are evidences of different data.

2.8 Differences for mock data

The mock data are fitted with the same pipeline up to and including `mass_total[1]`. Every stage up to that point has the same model, priors and sampler settings, so the two pipelines share those fits. The differences are:

- **No grid search.** `subhalo[1]` is not run, because on a simulation the subhalo's position is known.
- **Truth-centred subhalo prior.** In `subhalo[2]` the subhalo-centre priors are Gaussians centred on the true subhalo centre of the simulation ($(-0.10'', -1.90'')$, read from `info.json`), with $\sigma = 1.5''$ (`--centre_sigma`). This is the width of the real-data `subhalo[2]` prior with the 2×2 grid. M_{200}, the sampler settings ($n_{\mathrm{live}} = 600$, uncapped, `n_batch` $= 10$) and every other prior are unchanged.
- **One detection statistic**, $\Delta \ln Z = \ln Z_{\texttt{subhalo[2]}} - \ln Z_{\texttt{mass_total[1]}}$. Narrowing the centre prior by a factor two raises it by about $2 \ln 2 \approx 1.4$ while the prior is much wider than the posterior. This shift is the same in every band set, so the single- versus joint-band comparison does not depend on σ. The `no_subhalo` control constrains no centre and barely responds to σ, so it is compared with `subhalo_m10` at the same σ only. The values are not directly comparable with the real-data grid values.
- **Positions.** The multiple-image positions are solved from the true lens model of the simulation, so for `subhalo_m10` they include the subhalo's small image-position shift.

- **Data inspection.** The simulated data have no contaminants, and every band uses the simulation's $3.5''$ mask about $(0'', 0'')$. The inspection step for real data is not needed.

References

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